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Attention Mechanisms in Medical Image Analysis: Advance, Challenge, and Future Direction

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***Abstract*—Attention mechanisms have revolutionized medical image analysis by dynamically focusing on critical regions to address challenges like ambiguous lesion boundaries and high inter-class similarity. This paper explores their evolution from human visual attention research to hybrid architectures (such as channel-spatial attention and transformer), and focuses on their cutting-edge applications in tasks such as medical image segmentation, classification and enhancement. These mechanisms improve diagnostic accuracy by refining lesion delineation, highlighting discriminative features, and enabling multimodal fusion. Despite progress, challenges such as data scarcity, computational complexity, and cross-modal adaptability persist. Future work should prioritize lightweight designs, interpretability, and multimodal fusion to bridge the gap between algorithmic innovation and clinical deployment.**

***Index Terms*—Attention mechanisms, medical image analysis, segmentation, classification, Transformer**

I. Introduction

Driven by evolving computational demands, deep learning algorithms have also undergone rapid iteration and innovation, from traditional convolutional neural networks (CNN) [1], [2] to recurrent neural networks (RNN) [3], to generative adversarial networks (GAN) [4], [5] and Transformer architectures [6], each new algorithm injects new vitality into the development of AI. Under the continuous advancement of this profound revolution, the practical clinical application of deep learning technology in assisting medical image diagnosis has become increasingly common. Among them, cutting-edge technologies such as channel attention, hybrid attention, and Transformer have demonstrated excellent performance in the field of medical image processing [7], [8]. These technologies have powerful automatic feature extraction capabilities, which can accurately and efficiently extract key features from massive medical image data.

Medical image analysis faces challenges such as ambiguous lesion boundaries and high inter-class similarity. Attention mechanisms address these by dynamically focusing on critical regions, enabling more accurate diagnosis. This article reviews the latest progress of attention mechanisms in medical image analysis, focusing on their core concepts, applications in segmentation, classification, and generation tasks, and discussing open challenges.

II. Fundamental Concepts and Evolution of Attention Mechanisms

The concept of attention mechanism originated from the study of human visual attention process [9]. Inspired by the human visual system's ability to selectively focus on salient regions rather than processing all stimuli equally, attention mechanisms were developed to improve feature extraction in neural networks [10]–[13]. This ingenious mechanism has brought profound enlightenment to the research in the field of artificial intelligence, and attention mechanism has emerged [14]. Initially applied in NLP [15], attention mechanisms were later extended to computer vision and medical imaging, evolving into hybrid architectures (e.g., combining channel, spatial, and self-attention) [16]–[22].

Medical image analysis, as a special application scenario, has extremely high requirements for accurately identifying and analyzing lesion areas and key features in images. And the attention mechanism can precisely meet this requirement, as it can automatically identify and emphasize lesion areas or key features in medical images. Technologies such as channel attention, spatial attention, hybrid attention, and Transformer have demonstrated powerful capabilities in medical image

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processing. They can not only automatically extract data features, but also optimize the model's attention to key areas, improving diagnostic efficiency and accuracy. Channel and spatial attention modules have been widely explored—typical examples include squeeze-and-excitation (SE) [23] and efficient channel attention (ECA) [24] for channel modeling, and pyramid or convolution-based spatial attention for location refinement [25]–[29]. The Transformer architecture has achieved great success in the field of natural language processing, and its application to medical image analysis has brought new ideas and methods for the implementation of spatial attention [30], [31]. The hybrid attention mechanism cleverly combines multiple attention mechanisms, such as simultaneously fusing channel attention and spatial attention.

The development of attention mechanisms has gone through an evolution from traditional hybrid paradigms to multi-dimensional collaborative architectures. With the breakthrough progress of Transformer architecture, hybrid attention has begun to break through the binary boundaries between channels and space, forming a new paradigm by integrating the global modeling capability of self attention. For example, the cascading design of CBAM [32] and self attention modules (such as SCSA [33]) can simultaneously capture local channel associations and long-range spatial dependencies, while models such as HAT [34] further combine coordinate attention with sliding window self attention to achieve efficient localization of lesion areas in medical image segmentation.

III. Applications of Attention Mechanisms in Medical Imaging Tasks

In image segmentation tasks, attention mechanisms enhance segmentation by focusing on lesion boundaries and suppressing background noise, leading to improved Dice and IoU metrics (Table I). The values in the table are from the cited papers. For example, CASF-Net [35] uses a dual branch encoder to extract local details and global features using CNN and Transformer, respectively. Through a cross attention mechanism (DCF module), these two features are interactively fused during the encoding stage to enhance semantic expression ability, further enhancing the segmentation robustness of complex medical scenes such as glandular tissue and surrounding cell adhesions. It is worth noting that existing research has shown an evolutionary trend of "hierarchical attention structure": from the hybrid attention module proposed by Al et al. [36] to the channel space dual attention network constructed by DCA [37], researchers have gradually extended the attention mechanism from a single feature layer to cross layer interaction between codecs. This evolution is also reflected in TransAttUnet [38], which combines Transformer's self attention (TSA) with global spatial attention (GSA) through a self aware attention module (SAA), effectively capturing long-range dependencies and global spatial context information in medical images.

In classification tasks, attention modules highlight discriminative regions—such as lung nodules or lesions—by focusing on specific regions in medical images, the model can extract more discriminative features, thereby boosting accuracy and F1 scores across benchmarks (see Table II). The values in the table are from the cited papers. The current research presents two major technological routes. One technical route is that it is based on the CNN Transformer hybrid architecture for global local feature collaboration. For example, Wu et al. proposed CTransCNN [52], which adaptively captures implicit correlations between medical labels (such as the association between lung abnormalities and cardiac lesions) through a multi head attention module embedded in labels, and dynamically integrates CNN local features with Transformer global semantics using a cross branch information interaction module, effectively alleviating data imbalance problems (such as rare disease labels). Manzari et al. proposed MedViT [53], which generates attention weights through convolution operations, reducing computational complexity while preserving

TABLE I
Recent medical image segmentation methods and their performance metrics

Method	Year	Dataset	Performance(Dice/IoU)
DCA [37]	2023	GlaS	89.90%/81.68%
		MoNuSeg	79.50%/65.97%
		CVC- ClinicDB	90.86%/83.47%
		Kvasir-Seg	85.16%/74.34%
		Synapse	80.22%/69.80%
D-LKA Net [39]	2023	Synapse	84.27%/-
		ISIC 2017	92.54%/-
		ISIC 2018	91.77%/-
		PH²	94.90%/-
MERIT [40]	2023	Synapse	84.90%/-
		ACDC	92.32%/-
CRAPNet [41]	2023	Abdominal-CT	73.66%/-
		Abdominal-MRI	79.79%/-
H2Former [42]	2023	Kvasir-SEG	91.80%/86.29%
		ISIC 2016	92.41%/86.45%
		ACDC	92.40%/-
		Synapse	70.52%/-
FCT [43]	2023	ACDC	93.02%/-
		Synapse	83.53%/-
CaraNet [44]	2023	BraTS 2018	63.1%/-
		Kvasir-SEG	91.8%/-
		CVC-ColonDB	77.3%/-
		CVC-ClinicDB	93.6%/-
		CVC-300	90.3%/-
MedSegDiff [45]	2023	REFUGE-2	86.9%/78.5%
		BraTS-2021	89.9%/82.3%,
		DDTI	86.1%/79.6%
LViT [46]	2023	QaTa-COV19	83.66%/ 75.11%
		MosMedData+	74.57%/ 61.33%
		ESO-CT	71.53%/ 59.94%
FTransCNN [47]	2023	Chest X-ray	95.80%/ 93.03%
		Kvasir-SEG	88.69%/ 81.79%
MedNeXt [48]	2023	BTCV	84.82%/-
		AMOS22	91.77%/-
		KiTS19	90.71%/-
		BraTS21	88.01%/-
DCSAU-Net [49]	2023	CVC-ClinicDB	91.6%/86.1%
		ISIC 2018	90.4%/84.1%
		SegPC-2021	88.6%/80.6%
		BraTS-2021	78.8%/70.3%
MSDA-Net [50]	2024	CVC-ClinicDB	0.917/0.870
		ISIC 2018	91.3%/85.7%
Mamba-UNet [51]	2024	ACDC	92.81%/86.98%
		Synapse	64.29%/54.05%

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TABLE II
RECENT MEDICAL IMAGE CLASSIFICATION METHODS AND THEIR PERFORMANCE METRICS

Method	Year	Dataset	Performance(ACC/F1)
MedViT [53]	2023	MedMNIST-2D	84.2%/-
DiffMIC [54]	2023	PMG2000	93.1%/0.926
		HAM10000	90.6%/0.816
		APTOS2019	85.8%/0.716
SDDA-MAE [55]	2023	SIIM-ACR	94.64%/-
		BUSI	92.53%/-
FEB-GPB [56]	2023	DDR Dataset	82.35%/-
		IDRiD Dataset	80.58%/ 0.75
MACX [57]	2023	SARS-CoV-2 CT	96.99%/0.97
Med-Former [58]	2023	NIH Chest X-ray14	87.6%/-
		DermaMNIST	78.3%/-
		BloodMNIST	96.5%/-
AHANet [59]	2023	ADNI	98.53%/0.9853
UKSSL [60]	2023	LC25000	98.9%/98.9%
		BCCD	94.1%/0.943
DDMARN [61]	2023	Kidney Biopsy HSI	96.23%/-
SmAP [62]	2024	RSNA	-/0.787
		PANDA	-/0.915
		CAMELYON16	-/0.916
MedMamba-B [63]	2024	Fetal-Planes-DB	94.4%/-
		Otoscopy2024	89.9%/-
		MedMNIST-BloodMNIST	98.4%/-
MedViTV2-Large [64]	2024	PAD-UFES-20	63.6%/-
		ISIC2018	77.1%/-
		MedMNIST	98.7%/-
DCCAM [65]	2025	ADNI	98.40%/-

local contextual information; Simultaneously introducing an efficient self attention module (ESA) for multi-scale feature fusion. Especially in the classification task of chest X-ray pneumonia, this model effectively captures the local texture features of lesion areas through convolutional attention, and uses self attention to establish global correlations between different lung lobe lesions. Another technical route is dynamic attention optimization combined with other models such as generative models. For example, DiffMIC [54] adopts a strategy that combines self attention and spatial attention. This method combines the iterative denoising characteristics of the diffusion model with the attention mechanism. Through the global local two-way feature interaction, noise robustness and fine-grained feature capture are achieved in placental maturity classification, skin lesion classification and diabetes retinopathy classification tasks.

In other image processing tasks such as medical image enhancement and super-resolution, attention-based models improve multimodal fusion and detail reconstruction by selectively weighting informative anatomical and pathological regions [66]. The core value of attention mechanism lies in the dual constraint of maintaining anatomical structure and enhancing pathological features, Georgescu et al. [67] focused on medical image super-resolution and developed a multimodal multi head convolutional attention module (MMHCA), which jointly optimizes super-resolution through multimodal low resolution inputs. For the first time, medical super-resolution relies solely on multimodal low resolution inputs without the need for cross modal high-resolution guidance. By integrating multi-scale spatial information through multi head design, a new approach is provided for multimodal medical image enhancement. Song et al. [68] proposed an innovative method combining channel and spatial attention with dynamic convolution mechanism for medical image enhancement. The author designed the Cross Group Attention module to address the issue of spatial misalignment between modalities in multimodal MRI image synthesis, and introduced channel shuffle to achieve cross group information exchange, generating dynamic weights to suppress cross modal aliasing noise. Significantly improved the feature alignment and fusion performance of multimodal MRI such as T1/T2/PD under misalignment conditions. By innovatively designing the attention mechanism, these studies aim to solve the core challenges of multimodal data fusion, low-quality enhancement and dynamic modeling in medical imaging, realize the collaborative optimization of accurate anatomical structure maintenance and significant enhancement of pathological features, and ultimately improve the credibility and interpretability of clinical diagnosis [66], [69]–[71].

IV. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

Although attention mechanisms offer clear benefits, their clinical deployment faces key challenges. First, the scarcity and high annotation costs of medical data limit the generalization capabilities of models. Existing algorithms heavily rely on large-scale annotated datasets, yet acquiring medical images requires strict adherence to ethical and privacy protocols. Secondly, the diversity of medical imaging modalities (e.g., CT, MRI, ultrasound) and pathological complexity (e.g., variable lesion morphology, ambiguous tissue boundaries) impose higher demands on the feature selection capabilities of attention mechanisms. For instance, in cross-modal image fusion tasks, current methods often introduce noise due to spatial misalignment or resolution discrepancies between modalities, compromising reconstruction quality. Moreover, the computational complexity of attention mechanisms (e.g., self-attention in Transformers) may become a bottleneck in clinical scenarios requiring real-time performance, particularly when processing high-resolution 3D medical images, where memory and computational costs escalate significantly, hindering deployment efficiency.

Future research must seek breakthroughs in both algorithmic innovation and clinical adaptability. At the algorithmic level, lightweight and efficient designs are critical. For example, developing sparse attention mechanisms or hybrid local-global architectures could reduce computational overhead while retaining long-range dependency modeling. Exploring self-supervised or few-shot learning paradigms may alleviate reliance on annotated data. Additionally, enhancing multimodal collaboration and dynamic adaptability is essential. Cross-modal attention mechanisms could achieve semantic alignment of heterogeneous data, while meta-learning could

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dynamically adjust attention weights to improve the capture of complex pathological features. Clinically, model interpretability is paramount. Developing visualization tools to reveal correlations between attention-focused regions and diagnostic decisions can build trust among clinicians. Simultaneously, validating algorithms in real-world healthcare environments—for example, through federated learning frameworks that ensure data privacy in multi-center collaborative training—will enhance robustness across devices and populations.

The integration of emerging technologies injects new momentum into the field. For instance, synergies between diffusion models and attention mechanisms could enhance fidelity to pathological details in generative tasks. The introduction of state-space models (e.g., Mamba) may overcome Transformer's sequence-length limitations, improving efficiency in analyzing longitudinal medical images. Ultimately, the evolution of attention mechanisms must anchor clinical needs, balancing accuracy, efficiency, and interpretability to propel artificial intelligence from auxiliary diagnosis toward precision medicine.

V. Conclusion

In conclusion, attention mechanisms have substantially advanced medical image analysis by improving lesion localization in segmentation, enhancing discriminative feature learning in classification, and refining detail reconstruction in generative tasks. However, challenges remain in data scarcity, computational efficiency, and interpretability. Future work should focus on lightweight attention architectures, robust cross-modal fusion, and explainable attention visualizations to facilitate clinical translation.

References

[1] J. Gehring, M. Auli, D. Grangier, D. Yarats, and Y. N. Dauphin, "Convolutional sequence to sequence learning," in *International conference on machine learning*, pp. 1243–1252, PMLR, 2017.

[2] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 7794–7803, 2018.

[3] W. Zaremba, I. Sutskever, and O. Vinyals, "Recurrent neural network regularization," *arXiv preprint arXiv:1409.2329*, 2014.

[4] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial networks," *Communications of the ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[5] X. Li, Y. Jiang, J. J. Rodriguez-Andina, H. Luo, S. Yin, and O. Kaynak, "When medical images meet generative adversarial network: recent development and research opportunities," *Discover Artificial Intelligence*, vol. 1, pp. 1–20, 2021.

[6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[7] Q. Guan, Y. Huang, Z. Zhong, Z. Zheng, L. Zheng, and Y. Yang, "Diagnose like a radiologist: Attention guided convolutional neural network for thorax disease classification," *arXiv preprint arXiv:1801.09927*, 2018.

[8] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, *et al.*, "Attention u-net: Learning where to look for the pancreas," *arXiv preprint arXiv:1804.03999*, 2018.

[9] M. Corbetta and G. L. Shulman, "Control of goal-directed and stimulus-driven attention in the brain," *Nature reviews neuroscience*, vol. 3, no. 3, pp. 201–215, 2002.

[10] R. A. Rensink, "The dynamic representation of scenes," *Visual cognition*, vol. 7, no. 1-3, pp. 17–42, 2000.

[11] M. Hayhoe and D. Ballard, "Eye movements in natural behavior," *Trends in cognitive sciences*, vol. 9, no. 4, pp. 188–194, 2005.

[12] D. Noton and L. Stark, "Scanpaths in saccadic eye movements while viewing and recognizing patterns," *Vision research*, vol. 11, no. 9, pp. 929–IN8, 1971.

[13] J. Cai, Y. Li, B. Liu, Z. Wu, S. Zhu, Q. Chen, Q. Lei, H. Hou, Z. Guo, H. Jiang, S. Guo, F. Wang, S. Huang, S. Zhu, X. Fan, and S. Tao, "Developing deep lstms with later temporal attention for predicting covid-19 severity, clinical outcome, and antibody level by screening serological indicators over time," *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 7, pp. 4204–4215, 2024.

[14] X. Wang, R. Lv, Y. Zhao, T. Yang, and Q. Ruan, "Multi-scale context aggregation network with attention-guided for crowd counting," in *2020 15th IEEE International Conference on Signal Processing (ICSP)*, vol. 1, pp. 240–245, IEEE, 2020.

[15] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," *arXiv preprint arXiv:1409.0473*, 2014.

[16] J. Fu, J. Liu, H. Tian, Y. Li, Y. Bao, Z. Fang, and H. Lu, "Dual attention network for scene segmentation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 3146–3154, 2019.

[17] Y. Yuan, L. Huang, J. Guo, C. Zhang, X. Chen, and J. Wang, "Ocnet: Object context network for scene parsing," *arXiv preprint arXiv:1809.00916*, 2018.

[18] K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhudinov, R. Zemel, and Y. Bengio, "Show, attend and tell: Neural image caption generation with visual attention," in *International conference on machine learning*, pp. 2048–2057, PMLR, 2015.

[19] K. Gregor, I. Danihelka, A. Graves, D. Rezende, and D. Wierstra, "Draw: A recurrent neural network for image generation," in *International conference on machine learning*, pp. 1462–1471, PMLR, 2015.

[20] H. Zhang, I. Goodfellow, D. Metaxas, and A. Odena, "Self-attention generative adversarial networks," in *International conference on machine learning*, pp. 7354–7363, PMLR, 2019.

[21] Q. Wang, T. Wu, H. Zheng, and G. Guo, "Hierarchical pyramid diverse attention networks for face recognition," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8326–8335, 2020.

[22] J. Yang, P. Ren, D. Zhang, D. Chen, F. Wen, H. Li, and G. Hua, "Neural aggregation network for video face recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4362–4371, 2017.

[23] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 7132–7141, 2018.

[24] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "Eca-net: Efficient channel attention for deep convolutional neural networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 11534–11542, 2020.

[25] X. Hu, Z. Zhang, Z. Jiang, S. Chaudhuri, Z. Yang, and R. Nevatia, "Span: Spatial pyramid attention network for image manipulation localization," in *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXI 16*, pp. 312–328, Springer, 2020.

[26] M. Jaderberg, K. Simonyan, A. Zisserman, *et al.*, "Spatial transformer networks," *Advances in neural information processing systems*, vol. 28, 2015.

[27] V. Mnih, N. Heess, A. Graves, and K. Kavukcuoglu, "Recurrent models of visual attention," *Advances in neural information processing systems*, vol. 27, 2014.

[28] J. Hu, L. Shen, S. Albanie, G. Sun, and A. Vedaldi, "Gather-excite: Exploiting feature context in convolutional neural networks," *Advances in neural information processing systems*, vol. 31, 2018.

[29] T. Zhao and X. Wu, "Pyramid feature attention network for saliency detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 3085–3094, 2019.

[30] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[31] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted

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windows," in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 10012–10022, 2021.

[32] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "Cbam: Convolutional block attention module," in *Proceedings of the European conference on computer vision (ECCV)*, pp. 3–19, 2018.

[33] Y. Si, H. Xu, X. Zhu, W. Zhang, Y. Dong, Y. Chen, and H. Li, "Scsa: Exploring the synergistic effects between spatial and channel attention," *Neurocomputing*, vol. 634, p. 129866, 2025.

[34] X. Chen, X. Wang, J. Zhou, Y. Qiao, and C. Dong, "Activating more pixels in image super-resolution transformer," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 22367–22377, 2023.

[35] J. Zheng, H. Liu, Y. Feng, J. Xu, and L. Zhao, "Casf-net: Cross-attention and cross-scale fusion network for medical image segmentation," *Computer Methods and Programs in Biomedicine*, vol. 229, p. 107307, 2023.

[36] A. Al Qurri and M. Almekkawy, "Improved unet with attention for medical image segmentation," *Sensors*, vol. 23, no. 20, p. 8589, 2023.

[37] G. C. Ates, P. Mohan, and E. Celik, "Dual cross-attention for medical image segmentation," *Engineering Applications of Artificial Intelligence*, vol. 126, p. 107139, 2023.

[38] B. Chen, Y. Liu, Z. Zhang, G. Lu, and A. W. K. Kong, "Transattunet: Multi-level attention-guided u-net with transformer for medical image segmentation," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 8, no. 1, pp. 55–68, 2023.

[39] R. Azad, L. Niggemeier, M. Hüttemann, A. Kazerouni, E. K. Aghdam, Y. Velichko, U. Bagci, and D. Merhof, "Beyond self-attention: Deformable large kernel attention for medical image segmentation," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 1287–1297, 2024.

[40] M. M. Rahman and R. Marculescu, "Multi-scale hierarchical vision transformer with cascaded attention decoding for medical image segmentation," in *Medical Imaging with Deep Learning*, pp. 1526–1544, PMLR, 2024.

[41] H. Ding, C. Sun, H. Tang, D. Cai, and Y. Yan, "Few-shot medical image segmentation with cycle-resemblance attention," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 2488–2497, 2023.

[42] A. He, K. Wang, T. Li, C. Du, S. Xia, and H. Fu, "H2former: An efficient hierarchical hybrid transformer for medical image segmentation," *IEEE Transactions on Medical Imaging*, vol. 42, no. 9, pp. 2763–2775, 2023.

[43] A. Tragakis, C. Kaul, R. Murray-Smith, and D. Husmeier, "The fully convolutional transformer for medical image segmentation," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 3660–3669, 2023.

[44] A. Lou, S. Guan, H. Ko, and M. H. Loew, "Caranet: context axial reverse attention network for segmentation of small medical objects," in *Medical Imaging 2022: Image Processing*, vol. 12032, pp. 81–92, SPIE, 2022.

[45] J. Wu, R. Fu, H. Fang, Y. Zhang, Y. Yang, H. Xiong, H. Liu, and Y. Xu, "Medsegdiff: Medical image segmentation with diffusion probabilistic model," in *Medical Imaging with Deep Learning*, pp. 1623–1639, PMLR, 2024.

[46] Z. Li, Y. Li, Q. Li, P. Wang, D. Guo, L. Lu, D. Jin, Y. Zhang, and Q. Hong, "Lvit: language meets vision transformer in medical image segmentation," *IEEE transactions on medical imaging*, vol. 43, no. 1, pp. 96–107, 2023.

[47] W. Ding, H. Wang, J. Huang, H. Ju, Y. Geng, C.-T. Lin, and W. Pedrycz, "Ftranscnn: Fusing transformer and a cnn based on fuzzy logic for uncertain medical image segmentation," *Information Fusion*, vol. 99, p. 101880, 2023.

[48] S. Roy, G. Koehler, C. Ulrich, M. Baumgartner, J. Petersen, F. Isensee, P. F. Jaeger, and K. H. Maier-Hein, "Mednext: transformer-driven scaling of convnets for medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pp. 405–415, Springer, 2023.

[49] Q. Xu, Z. Ma, W. Duan, *et al.*, "Dcsau-net: A deeper and more compact split-attention u-net for medical image segmentation," *Computers in Biology and Medicine*, vol. 154, p. 106626, 2023.

[50] C. Zhu, K. Cheng, and X. Hua, "A medical image segmentation network with multi-scale and dual-branch attention," *Applied Sciences*, vol. 14, no. 14, p. 6299, 2024.

[51] Z. Wang, J.-Q. Zheng, Y. Zhang, G. Cui, and L. Li, "Mamba-unet: Unet-like pure visual mamba for medical image segmentation," *arXiv preprint arXiv:2402.05079*, 2024.

[52] X. Wu, Y. Feng, H. Xu, Z. Lin, T. Chen, S. Li, S. Qiu, Q. Liu, Y. Ma, and S. Zhang, "Ctranscnn: Combining transformer and cnn in multilabel medical image classification," *Knowledge-Based Systems*, vol. 281, p. 111030, 2023.

[53] O. N. Manzari, H. Ahmadabadi, H. Kashiani, S. B. Shokouhi, and A. Ayatollahi, "Medvit: a robust vision transformer for generalized medical image classification," *Computers in biology and medicine*, vol. 157, p. 106791, 2023.

[54] Y. Yang, H. Fu, A. I. Aviles-Rivero, C.-B. Schönlieb, and L. Zhu, "Diffmic: Dual-guidance diffusion network for medical image classification," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pp. 95–105, Springer, 2023.

[55] Y. Wang, S. Chen, T. Wang, and J. Liu, "Sdda-mae: Self-distillation enhanced dual attention masked autoencoder for small-scale medical image datasets," in *Medical Imaging with Deep Learning*, 2024.

[56] Z. Gu, Y. Li, Z. Wang, J. Kan, J. Shu, and Q. Wang, "Classification of diabetic retinopathy severity in fundus images using the vision transformer and residual attention," *Computational Intelligence and Neuroscience*, vol. 2023, no. 1, p. 1305583, 2023.

[57] S. Ghosh and A. Chatterjee, "Automated covid-19 ct image classification using multi-head channel attention in deep cnn," *arXiv preprint arXiv:2308.00715*, 2023.

[58] G. J. Chowdary and Z. Yin, "Med-former: A transformer based architecture for medical image classification," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pp. 448–457, Springer, 2024.

[59] T. Illakiya, K. Ramamurthy, M. Siddharth, R. Mishra, and A. Udainiya, "Ahanet: Adaptive hybrid attention network for alzheimer's disease classification using brain magnetic resonance imaging," *Bioengineering*, vol. 10, no. 6, p. 714, 2023.

[60] Z. Ren, X. Kong, Y. Zhang, and S. Wang, "Ukssl: Underlying knowledge based semi-supervised learning for medical image classification," *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 5, pp. 459–466, 2023.

[61] X. Zhang, W. Li, C. Gao, Y. Yang, and K. Chang, "Hyperspectral pathology image classification using dimension-driven multi-path attention residual network," *Expert Systems with Applications*, vol. 230, p. 120615, 2023.

[62] F. M. Castro-Macías, P. Morales Alvarez, Y. Wu, R. Molina, and A. Katsaggelos, "Sm: enhanced localization in multiple instance learning for medical imaging classification," *Advances in Neural Information Processing Systems*, vol. 37, pp. 77494–77524, 2024.

[63] Y. Yue and Z. Li, "Medmamba: Vision mamba for medical image classification," *arXiv preprint arXiv:2403.03849*, 2024.

[64] O. N. Manzari, H. Asgariandehkordi, T. Koleilat, Y. Xiao, and H. Rivaz, "Medical image classification with kan-integrated transformers and dilated neighborhood attention," *arXiv preprint arXiv:2502.13693*, 2025.

[65] S. Muksimova, S. Umirzakova, N. Iskhakova, A. Khaitov, and Y. Im Cho, "Advanced convolutional neural network with attention mechanism for alzheimer's disease classification using mri," *Computers in Biology and Medicine*, vol. 190, p. 110095, 2025.

[66] J. Zhang, Y. Niu, Z. Shangguan, W. Gong, and Y. Cheng, "A novel denoising method for ct images based on u-net and multi-attention," *Computers in Biology and Medicine*, vol. 152, p. 106387, 2023.

[67] M.-I. Georgescu, R. T. Ionescu, A.-I. Miron, O. Savencu, N.-C. Ristea, N. Verga, and F. S. Khan, "Multimodal multi-head convolutional attention with various kernel sizes for medical image super-resolution," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 2195–2205, 2023.

[68] T. Song, Y. Wu, M. Hu, X. Luo, L. Wei, G. Wang, Y. Guo, F. Xu, and S. Zhang, "Cross group attention and group-wise rolling for multimodal medical image synthesis," *arXiv preprint arXiv:2411.14684*, 2024.

[69] O. F. Atli, B. Kabas, F. Arslan, A. C. Demirtas, M. Yurt, O. Dalmaz, and T. Cukur, "I2i-mamba: Multi-modal medical image synthesis via selective state space modeling," *arXiv preprint arXiv:2405.14022*, 2024.

[70] J. S. Yoon, C. Zhang, H.-I. Suk, J. Guo, and X. Li, "Sadm: Sequence-aware diffusion model for longitudinal medical image generation," in *International Conference on Information Processing in Medical Imaging*, pp. 388–400, Springer, 2023.

[71] D. Y. Lee, J. Y. Kim, and S. Y. Cho, "Improving medical image quality using a super-resolution technique with attention mechanism," *Applied Sciences*, vol. 15, no. 2, p. 867, 2025.

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